



Darshan
UNIVERSITY

Python Programming - 2301CS404

Lab - 14

Roll No.: 459

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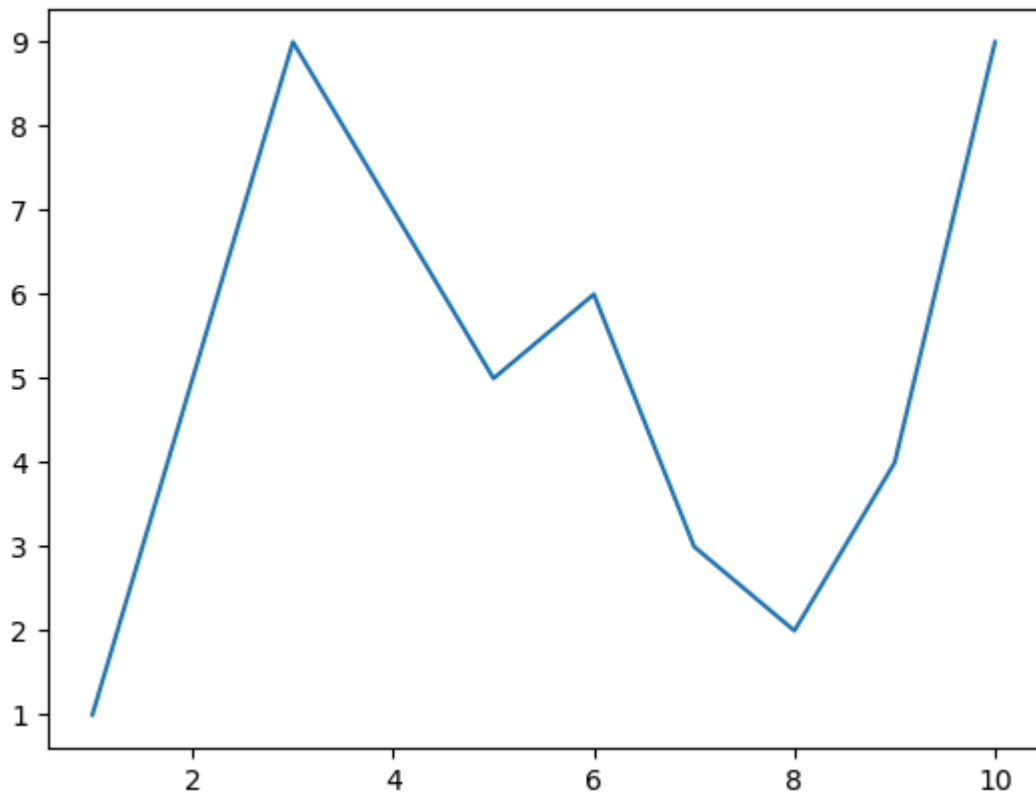
Import matplotlib below

```
In [2]: import matplotlib.pyplot as plt
```

01) WAP to display the line chart for given data for x and y axis.

```
In [ ]: x = range(1,11)  
        y = [1,5,9,7,5,6,3,2,4,9]
```

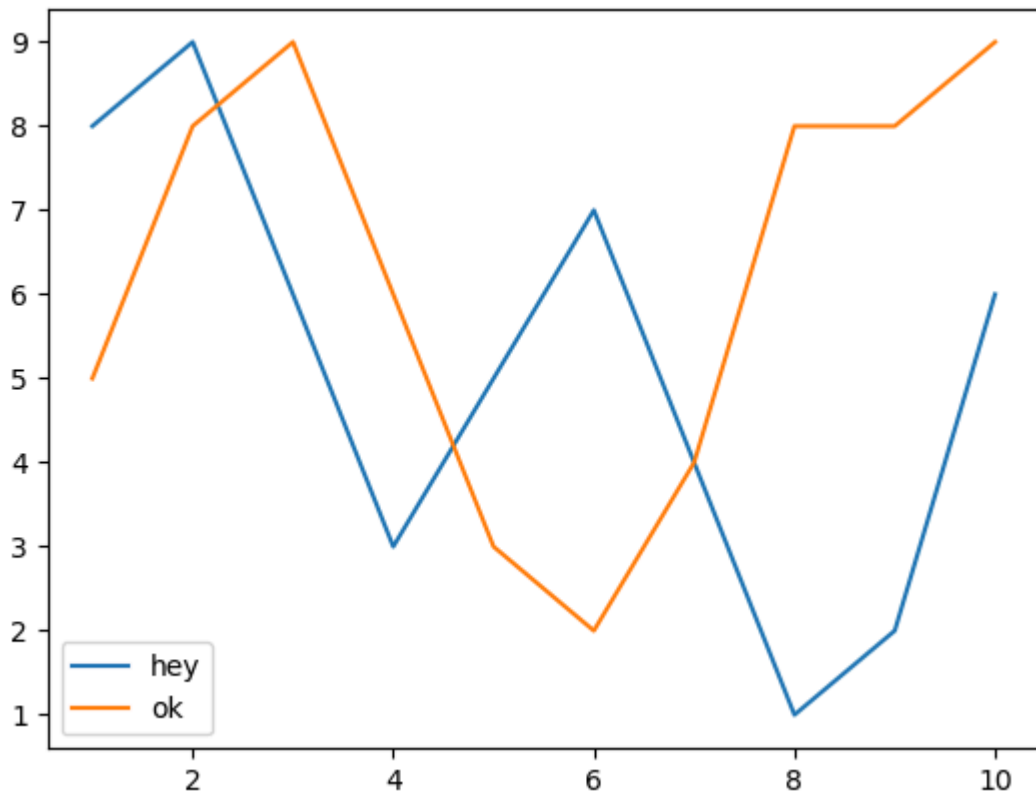
```
In [ ]: plt.plot(x,y)  
        plt.show()
```



02) WAP to generate line chart for given data.

```
In [ ]: x = range(1,11,1)
        cxMarks= [8,9,6,3,5,7,4,1,2,6]
        cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

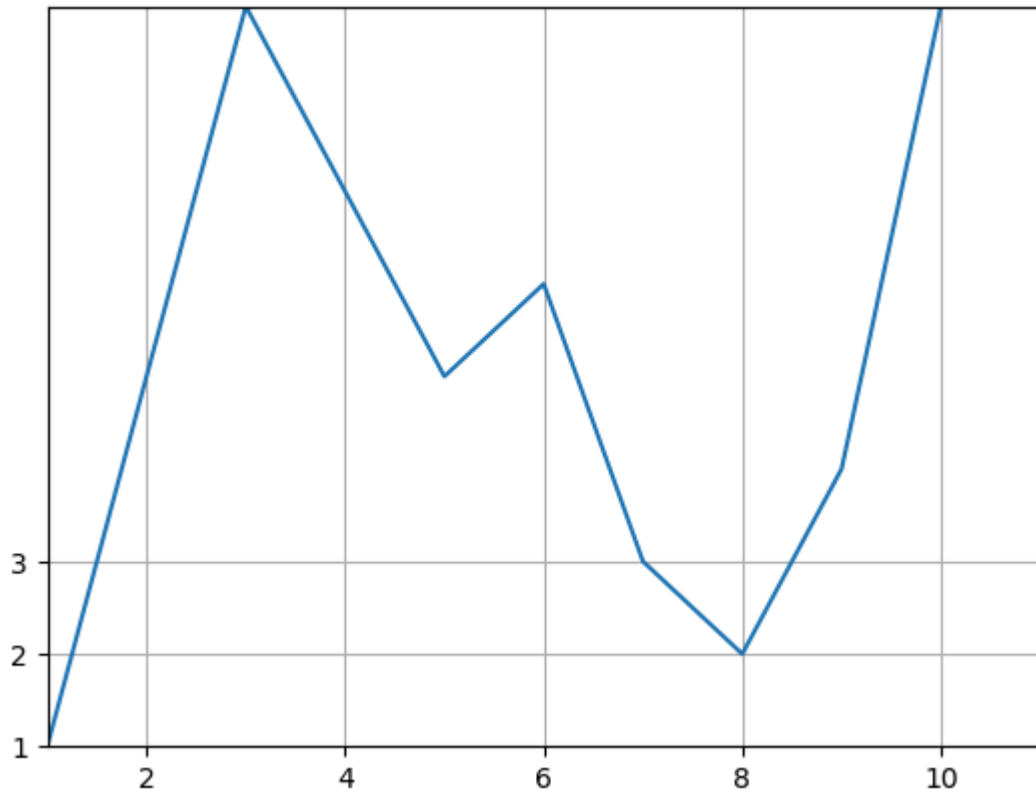
```
In [ ]: plt.plot(x,cxMarks,label="hey")
        plt.plot(x,cyMarks,label="ok")
        plt.legend()
        plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

```
In [ ]: x = range(1,11)
        y = [1,5,9,7,5,6,3,2,4,9]
```

```
In [ ]: ax = plt.axes()
        ax.set_xlim(1,11)
        ax.set_ylim(1,9)
        # ax.set_xticks(range(1,5))
        ax.set_yticks(range(1,4))
        ax.grid()
        plt.plot(x,y)
        plt.show()
```



04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperature.

```
In [ ]: Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
Temperature = [20, 32, 31, 33, 34, 43, 36]
```

```
In [ ]: maxVal = max(Temperature)
maxDay = Days[Temperature.index(maxVal)]

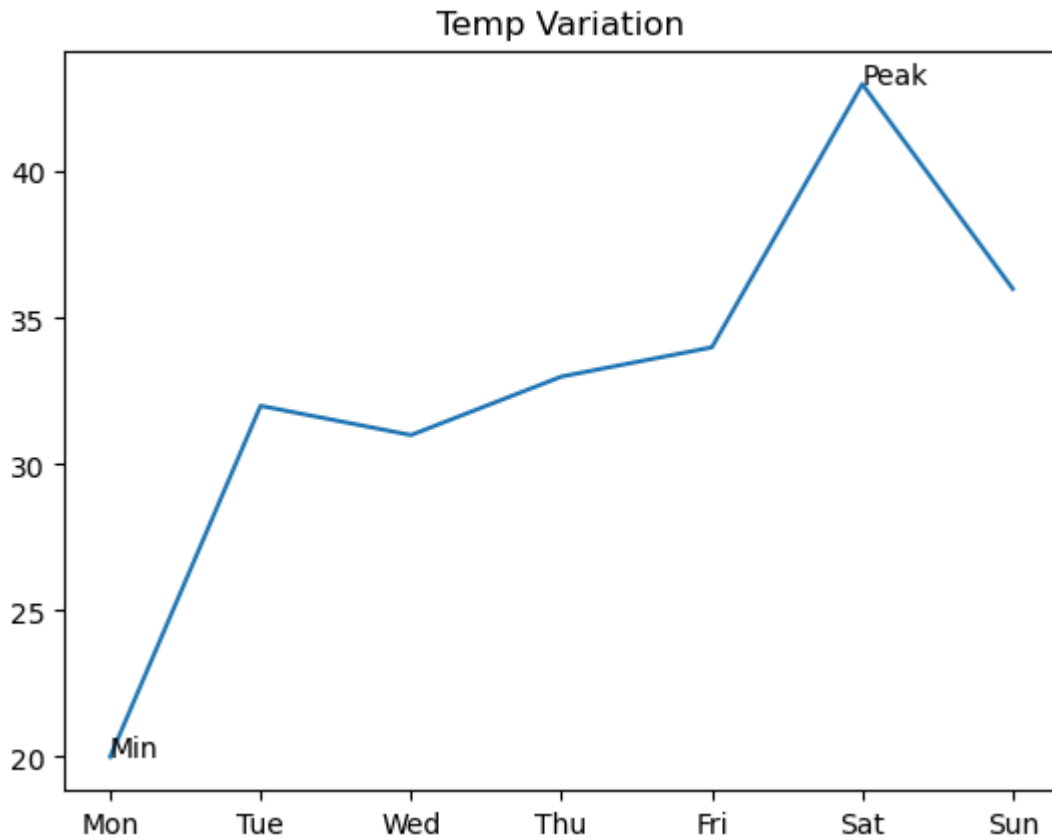
minVal = min(Temperature)
minDay = Days[Temperature.index(minVal)]

plt.plot(Days, Temperature)
plt.title("Temp Variation")

plt.annotate("Peak", xy=[maxDay, maxVal])
plt.annotate("Min", xy=[minDay, minVal])

# or

# plt.annotate('Shifted', xy=("Sat", 43), xytext=(3, 40),
#               arrowprops=dict(arrowstyle="->"))
plt.show()
```

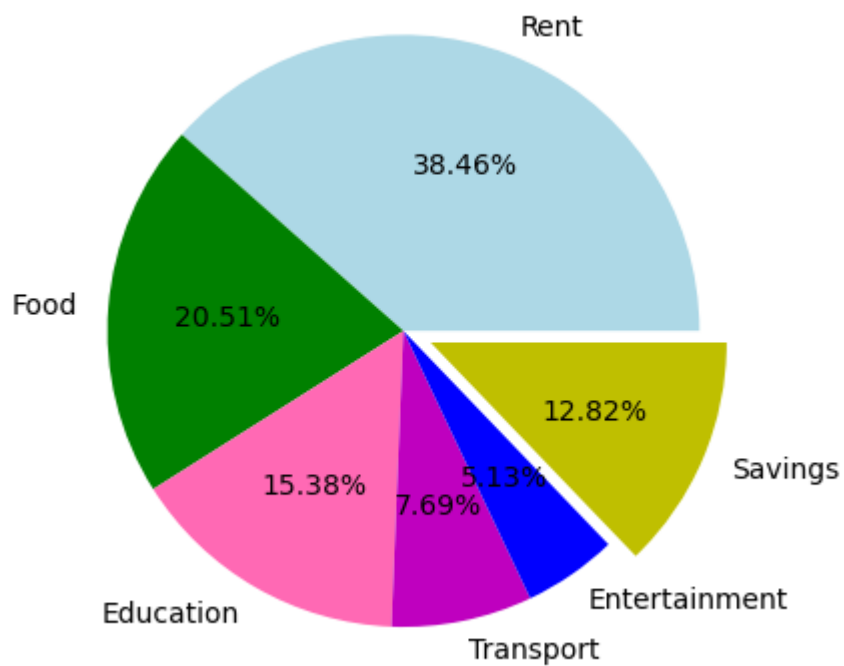


05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

```
In [ ]: Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Saving"]  
Amount = [15000, 8000, 6000, 3000, 2000, 5000]
```

```
In [ ]: plt.pie(Amount, labels=Categories, autopct = "%1.2f%%", colors = ["lightblue",  
plt.show()
```

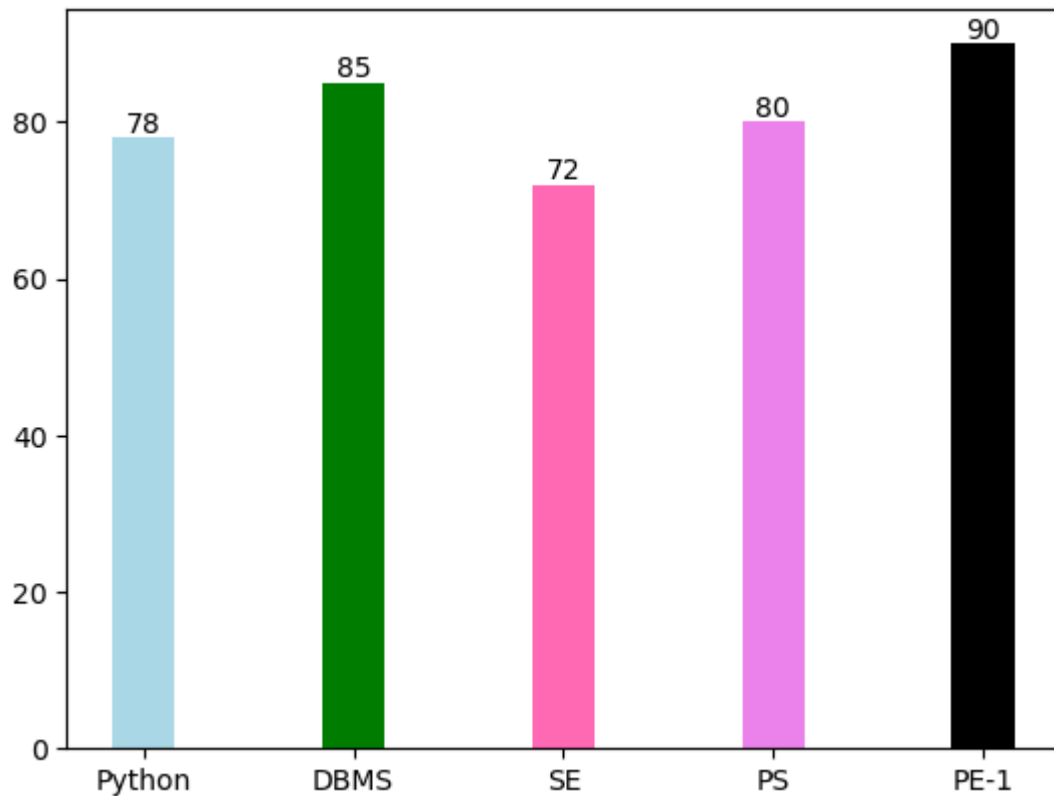


06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

```
In [ ]: Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]  
Marks = [78, 85, 72, 80, 90]
```

```
In [ ]: bars = plt.bar(Subjects,Marks,width=0.3,color = ["lightblue","g","hotpink","viol  
plt.bar_label(bars)  
plt.show()
```

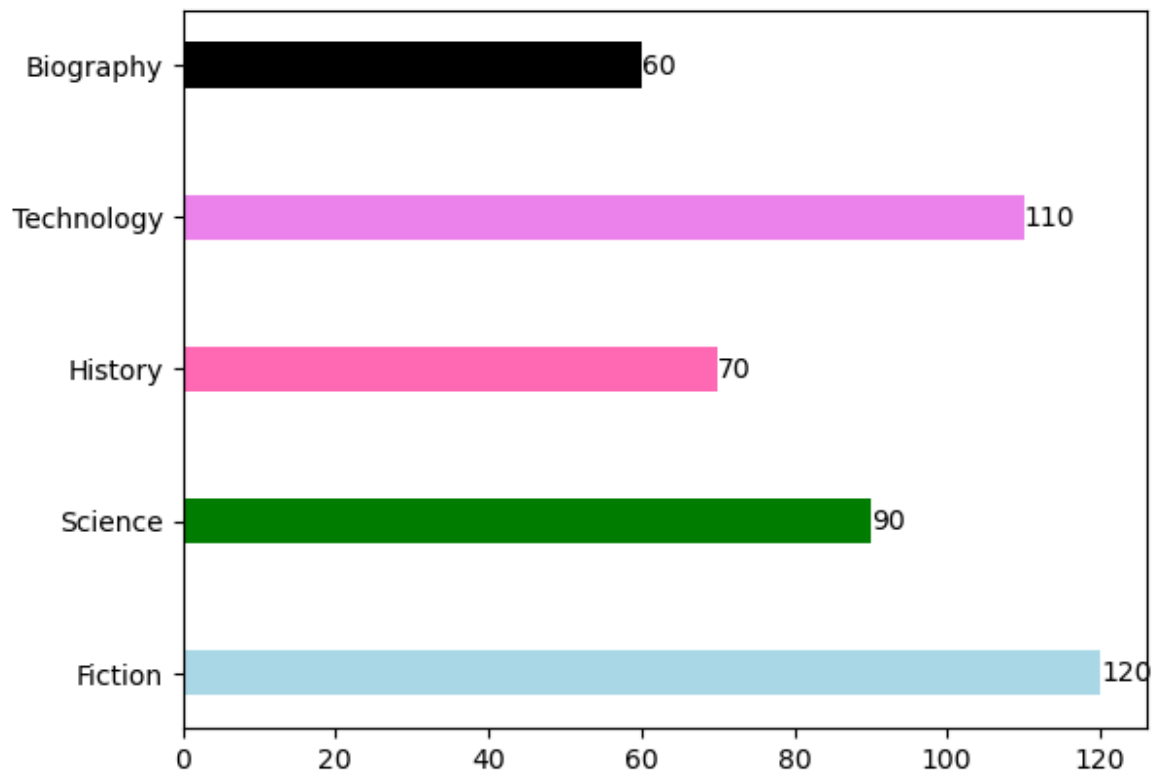


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

```
In [3]: Categories = ["Fiction", "Science", "History", "Technology", "Biography"]  
BooksSold = [120, 90, 70, 110, 60]
```

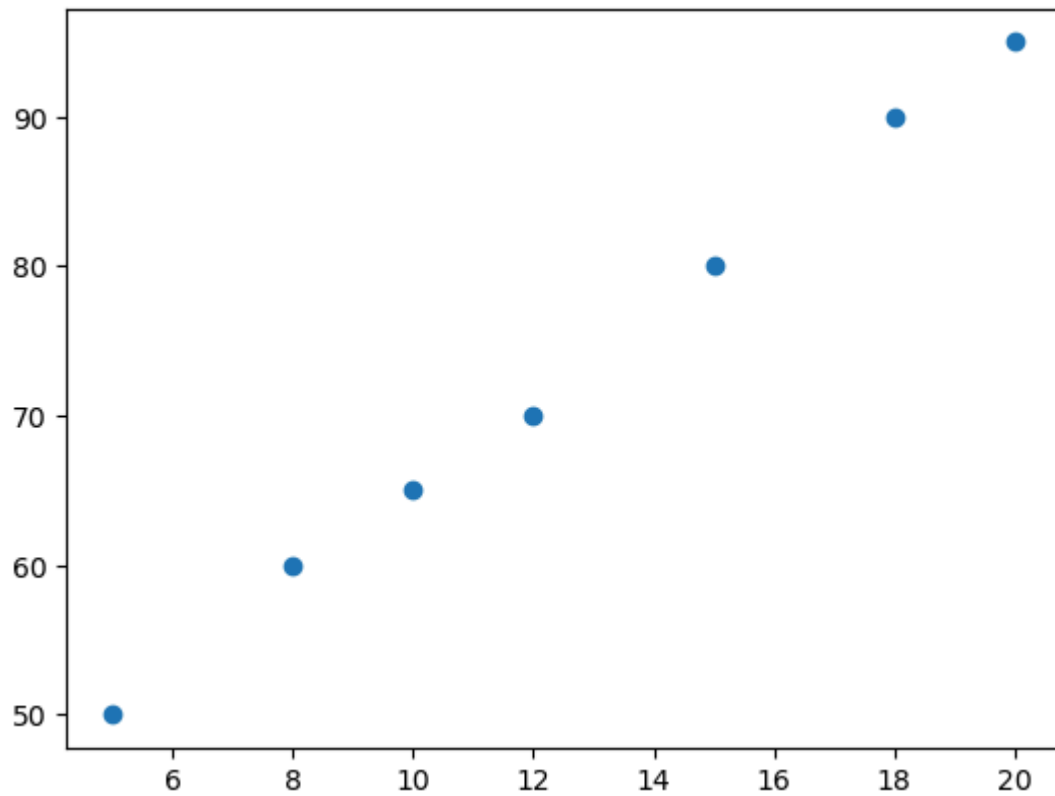
```
In [8]: bars = plt.barh(Categories, BooksSold, height=0.3, color = ["lightblue", "g", "hotpin  
plt.bar_label(bars)  
plt.show()
```



08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.

```
In [4]: Advertising = [5, 8, 10, 12, 15, 18, 20]
SalesRevenue = [50, 60, 65, 70, 80, 90, 95]
```

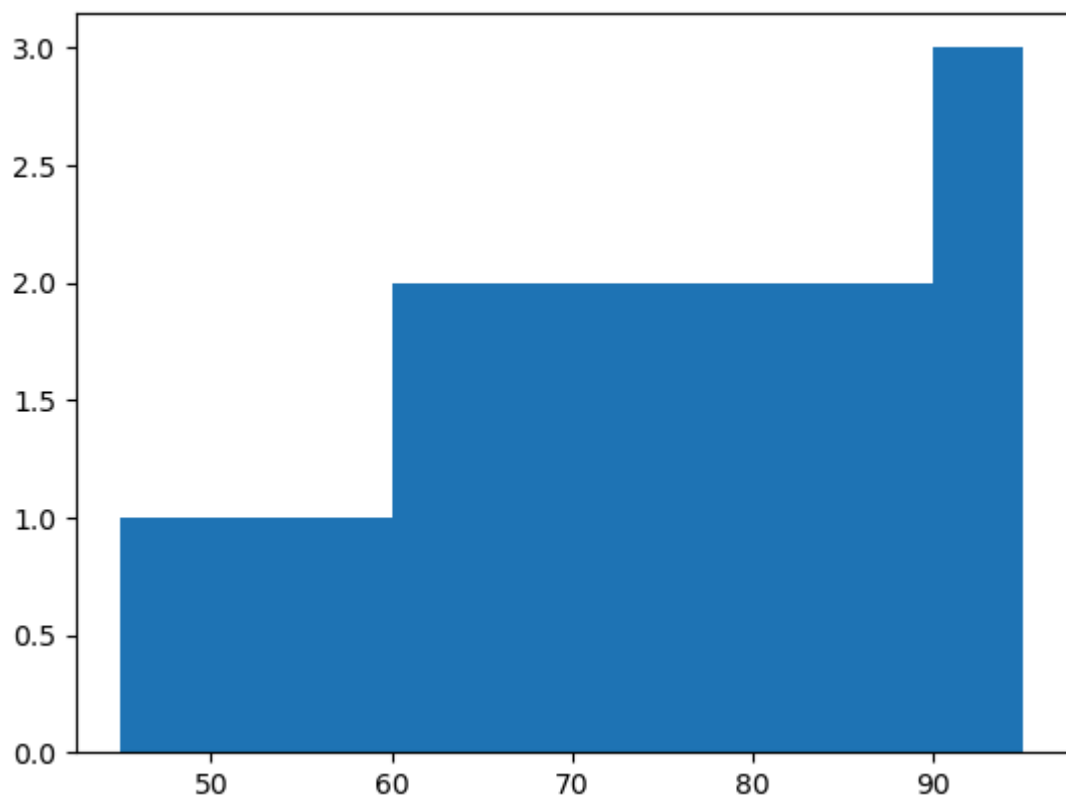
```
In [5]: plt.scatter(Advertising, SalesRevenue)
plt.show()
```

09) WAP to create a histogram representing the distribution of marks obtained by students in a class.

```
In [6]: Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

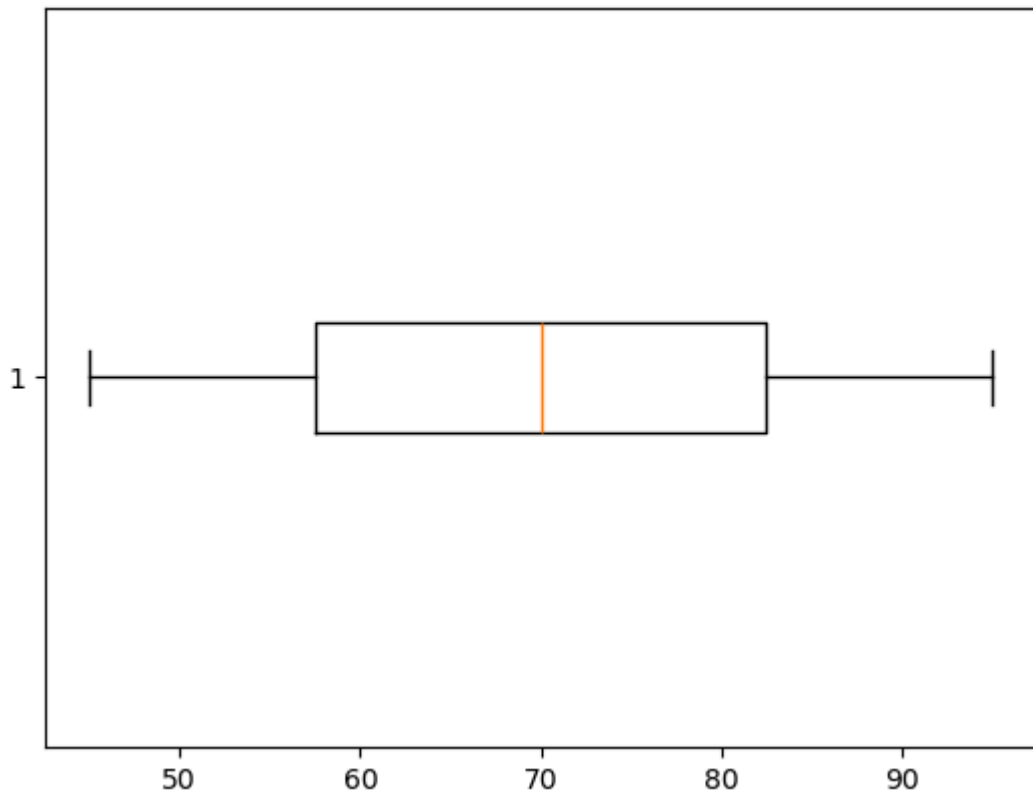
```
In [7]: plt.hist(Marks)
plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

```
In [15]: CommutingTime = [45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]
```

```
In [18]: plt.boxplot(CommutingTime, vert=False)  
plt.show()
```



```
In [ ]:
```